NISANTH K

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PROFESSIONAL SUMMARY

B.Tech Artificial Intelligence and Data Science student at Indra Ganesan College of Engineering with hands-on project experience across AI/ML, robotics, embedded hardware and full-stack web development. Builds practical systems that combine intelligent software, hardware and real-world problem solving, and has won first place at both a paper presentation and a project expo in 2026.

EDUCATION

B.Tech – Artificial Intelligence and Data Science *2025 – 2029 (Ongoing)*
Indra Ganesan College of Engineering

Higher Secondary – Computer Science Group *2023 – 2025 | 60%*
Vinmathee Matric Higher Secondary School

TECHNICAL SKILLS

Programming Languages: Python, C, SQL

Data Analysis: Pandas

Web Development: React, FastAPI, Django, MySQL

Embedded & Robotics: Jetson Nano, ESP32, PCA9685 servo driver, IMU sensor integration

AI Development Tools: Google Antigravity, Claude

Areas of Interest: Robotics, Artificial Intelligence, Machine Learning, Computer Vision, Embedded Systems, Intelligent Automation

FEATURED PROJECTS

Humanoid Robot (AURA) | *12-DOF humanoid* | *Jetson Nano, ESP32, PCA9685*

- Designing a 12-servo humanoid robot (MG995) with a split architecture: Jetson Nano for vision, speech and high-level behaviour; ESP32 for IMU reading and servo control through a PCA9685 driver.
- Hardware stack includes camera, microphone, speaker, display and gyroscope IMU; upper body (arms, torso, head) built, with standing-pose servo calibration done for all 12 joints.

Smart Gate Pass System with QR Code | *Web application* | *Django*

- Digitizes college gate-pass requests with role-based access for HOD, Student and Security, replacing handwritten records.
- A QR code is generated once the HOD approves a pass; security scans it to verify details and record the exit time.

Smart Attendance System (SmartAttend AI) | *React, FastAPI, SQLAlchemy, MySQL*

- Designed an attendance system on the department's existing MySQL database, with AI face recognition for theory periods and manual marking for labs.
- Role-based access for Admin, Faculty and Student; face data is designed to be consent-based and encrypted, with duplicate attendance prevented per session.

Smart Energy Generation System | *Energy monitoring*

- Explores technology-based energy generation and monitoring, with a focus on energy awareness, storage and efficient use.

ACHIEVEMENTS & CERTIFICATES

- **1st Prize** – CARE Tech Fest '26, Paper Presentation, CARE College of Engineering, Tiruchirappalli.
- **1st Place** – Project Expo 2026, Engineers' Day Celebrations, Indra Ganesan College of Engineering (15 September 2026).
- **Certificate of Appreciation** – WARTECH '26, Idea Presentation, Parisutham Institute of Technology & Science (28 August 2026).
- **2nd Place** – Esports (Free Fire), AIROX 2026 National-Level Technical Symposium, J.J. College of Engineering and Technology (22 August 2026).